

Research Profile

Researcher and engineer studying how generative models learn expressive representations, reason across modalities, and operate efficiently at scale. Research spans speech, sound, music, and broader multimodal intelligence. First- and co-first-author work has appeared at NeurIPS, ICML, ICLR, ACL, EMNLP, and ACM Multimedia, with **1,000+ Google Scholar citations**. Research systems have supported **100K+ hours** of training data, **5–10×** faster production inference, and widely adopted open-source releases.

Research Interests

Generative modeling · Multimodal representation learning · Cross-modal reasoning · Autoregressive models · Flow matching and Rectified Flow · Reinforcement learning · Speech and audio intelligence · Efficient training and inference

Research & Systems Experience

- Led research and engineering across model architecture, large-scale training, evaluation, streaming inference, and open-source release for speech and audio foundation models in an industrial research environment.
- Designed and deployed a Rectified Flow inference path for **CosyVoice 2**, improving production inference speed by **5–10×**.
- Architected a distributed training stack for **100K+ hours** of multi-domain speech, adopted across large-scale speech/audio models and cloud workloads.
- Led **ThinkSound**, including multimodal-LLM AudioCoT planning, the AudioCoT dataset, model training, evaluation, and the full open-source training and inference release.

Selected Research Highlights

AudioCALM: Continuous Autoregressive Language Modeling for Universal Audio Preprint · 2026
Replaces discrete-token prediction with flow matching over continuous audio latents; block-causal AR-Flow and asymmetric modality experts unify streaming speech, sound, and music generation in one autoregressive backbone.

STAR-VAE: Structured Topology-Aware Regularization for Audio ICML · 2026
Formalizes the Rate–Distortion–Regularity Trilemma and introduces channel-indexed regularization with a CNN–Mamba backbone, achieving state-of-the-art reconstruction and generation.

PrismAudio: Reasoning and Reinforcement Learning for Video-to-Audio ICLR · 2026
Decomposes semantic, temporal, aesthetic, and spatial reasoning into specialized chains of thought and rewards; Fast-GRPO makes multi-objective post-training practical.

ThinkSound: Chain-of-Thought Reasoning for Audio Generation and Editing NeurIPS · 2025
Introduces AudioCoT planning for any-to-audio generation, object-centric refinement, and language-guided editing in a unified open-source framework.

FlashAudio: Rectified Flows for Ultra-Fast Text-to-Audio ACL Oral · SAC Highlight · 2025
Introduces Rectified Flow for text-to-audio and enables single-step synthesis at **400× real time** while outperforming strong diffusion baselines.

AudioLCM: Latent Consistency Models for Text-to-Audio ACM Multimedia · 2024
Achieves high-quality generation in two to four steps at **333× real time**; the release became a widely used research baseline.

Education

Zhejiang University Sep 2021 – Mar 2024
M.S. in Artificial Intelligence · Top 0.1% of program

Shandong University Sep 2016 – Jun 2020
B.S. in Electronic Information

Selected First- and Co-First-Author Publications

2026	AudioCALM: Continuous Autoregressive Language Modeling for Universal Audio Generation	Preprint
	Huadai Liu et al. · Universal audio generation	
2026	STAR-VAE: Structured Topology-Aware Regularization for Audio Reconstruction and Generation	ICML
	Huadai Liu et al. · Audio representation learning	
2026	PrismAudio: Decomposed Chain-of-Thought and Multi-dimensional Rewards for V2A Generation	ICLR
	Huadai Liu et al. · Multimodal reasoning and RL	
2025	ThinkSound: Chain-of-Thought Reasoning in Multimodal LLMs for Audio Generation and Editing	NeurIPS
	Huadai Liu et al. · Audio reasoning	
2025	OmniAudio: Generating Spatial Audio from 360-Degree Video	ICML
	Huadai Liu et al. · Spatial audio generation	
2025	FlashAudio: Rectified Flows for Fast and High-Fidelity Text-to-Audio Generation	ACL
	Huadai Liu et al. · Oral · SAC Highlight	
2025	MEDIC: Zero-shot Music Editing with Disentangled Inversion Control	ACM MM
	Huadai Liu et al. · Oral	
2024	AudioLCM: Efficient and High-Quality Text-to-Audio Generation with Minimal Inference Steps	ACM MM
	Huadai Liu et al. · Efficient generation	
2024	Wav2SQL: Direct Generalizable Speech-To-SQL Parsing	ACL Findings
	Huadai Liu et al. · Speech reasoning	
2023	ViT-TTS: Visual Text-to-Speech with Scalable Diffusion Transformer	EMNLP
	Huadai Liu et al. · Multimodal TTS	

Selected collaborative work: SpaceVista (ICML 2026), MME-Emotion (ICLR 2026), MaskAudioFlow (Findings of ACL 2026), and one co-authored paper at ECCV 2026. Full and current record: Google Scholar.

Open Source & Research Leadership

22.5K+	CosyVoice	Core contributor to the official ecosystem; 2.6K forks. Designed the production Rectified Flow inference path for CosyVoice 2.
1.4K+	ThinkSound	Project lead; released the AudioCoT dataset, model weights, training and inference code, and demos. 81 forks across the official repository.
1.2K+	AudioLCM	Project lead; released the research implementation and established a widely used baseline for efficient audio generation.

Repository counts are rounded public snapshots as of August 2026.

Academic Service

Leadership	Area Chair, ACL Rolling Review (2026)
Recognition	Outstanding Reviewer, ECCV 2026
Conference reviewer	ICML · NeurIPS · ICLR · AAAI · ACL · EMNLP · ECCV · ACM Multimedia
Journal reviewer	IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)

Selected Honors

ECCV Outstanding Reviewer (2026) · ACL SAC Highlight Award (2025) · National Scholarship, top 1% (2023) · Outstanding Graduate of Zhejiang Province and Zhejiang University (2024)

Technical Expertise

Research: Generative modeling, speech synthesis, audio generation, video-to-audio, multimodal LLMs, diffusion models, Rectified Flow, flow matching, autoregressive models, reinforcement learning, GRPO, and chain-of-thought reasoning.
Engineering: Python, PyTorch, distributed training, FSDP, DeepSpeed, large-scale data pipelines, streaming inference, model optimization, production deployment, and C++ (working proficiency).
Languages: Chinese (native), English (professional).